Lokesh Setty

AI/ML ENGINEER

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Professional Summary

Results-driven **AI/ML Engineer and Data Scientist** with **4+ years of experience** in developing and deploying end-to-end Machine Learning, Generative AI, and data-driven solutions. Skilled in **Python, SQL, Scikit-learn, TensorFlow, PyTorch, FastAPI, LangChain, PySpark, MLflow, Docker, Kubernetes, AWS, Azure, Power BI, and Tableau**, with expertise in predictive modeling, deep learning, NLP, data analysis, MLOps, and scalable cloud-based AI/ML solutions.

Skills

Methodologies:	SDLC, Agile, Waterfall
Programming Language:	Python, SQL, R
Packages:	NumPy, Pandas, Matplotlib, SciPy, Scikit-learn, TensorFlow, Seaborn, ggplot2, Plotly, Keras, OpenCV, NLTK
ML/Generative AI:	Regression analysis, Bayesian Method, Decision Tree, Random Forests, Support Vector Machine, Neural Network, Sentiment Analysis, K-Means Clustering, KNN, Natural Language Processing (NLP), LangChain, OpenAI API, Hugging Face Transformers, Prompt Engineering, RAG Architecture, LLM Workflows, Semantic Search, Embeddings, Vector Databases (FAISS, Pinecone), LoRA, PEFT, instruction tuning
Visualization Tools:	Tableau, Power BI, Advanced Excel (Pivot Tables, VLOOKUP)
IDEs:	Visual Studio Code, PyCharm, Jupyter Notebook, IntelliJ
Cloud Technologies:	Amazon Web Services (AWS S3, EMR, Redshift), Microsoft Azure
Database:	MySQL, Oracle DB, PostgreSQL, MongoDB
Other Technical Skills:	Predictive Analysis, Time Series Forecasting, Regression Analysis, Hypothesis Testing, Classification, Clustering, Principal Component Analysis (PCA), Deep Learning, Model training and evaluation, SSIS, SSRS, Hadoop, Spark, Kafka, Snowflake, Big Query, Apache Airflow, Informatica, Talend, Jenkins, Kubernetes, Docker, Git

Work Experience

AI/ML Engineer | Morgan Stanley, USA — Jul 2024 - Present

- Developed autonomous **AI Agents** using **Python, LangGraph, LangChain, CrewAI, and OpenAI GPT-4** to automate investment research, financial analysis, and compliance workflows, reducing manual effort by **65%**.
- Designed and implemented **Retrieval-Augmented Generation (RAG)** pipelines using **Pinecone, FAISS, Sentence Transformers, and OpenAI Embeddings**, enabling semantic search across **10M+** financial documents and improving retrieval accuracy by **42%**.
- Integrated **OpenAI GPT-4, Anthropic Claude, and AWS Bedrock** APIs to build AI-powered assistants for financial document summarization, portfolio analysis, regulatory compliance, and intelligent question answering, improving response relevance by **40%**.
- Created scalable **FastAPI** microservices using **Python, Pydantic, AsyncIO, and REST APIs**, supporting over **3M AI inference requests/day** while reducing API response latency by **45%** through asynchronous processing.
- Built **multi-agent orchestration** workflows using **LangGraph** and **CrewAI**, enabling planner, researcher, compliance, and summarization agents to collaboratively execute complex financial workflows with dynamic task routing and shared memory.
- Implemented **long-term memory and contextual retrieval** using **Pinecone, Redis, and RAG**, enhancing contextual understanding and reducing LLM hallucinations by **55%** through intelligent knowledge retrieval.
- Automated the end-to-end **MLOps** lifecycle using **MLflow, Kubeflow, GitHub Actions, Docker, and Kubernetes**, streamlining model training, deployment, versioning, and monitoring while reducing deployment time by **80%**.
- Deployed cloud-native AI services on **AWS** using **Amazon SageMaker, EKS, S3, Lambda, Bedrock, IAM, and CloudWatch**, achieving **99.95%** platform availability with secure, auto-scalable infrastructure.
- Implemented **AI guardrails**, prompt validation, calibration scoring, and observability using **Prometheus, Grafana, ELK Stack, and FastAPI middleware**, reducing unsafe AI responses by **90%** and ensuring compliance with enterprise security and governance standards.

Data Scientist | Capgemini, India — Jan 2021 - Jul 2023

- Performed large-scale **data extraction, cleansing, validation, feature engineering, and exploratory data analysis (EDA)** using **Python, Pandas, NumPy, PySpark, and SQL**, improving data quality by **35%**.
- Developed supervised machine learning models using **XGBoost, Random Forest, Logistic Regression, Decision Trees, SVM, and LightGBM** to predict customer churn, purchase propensity, and sales performance, improving prediction accuracy by **30%**.
- Designed time-series forecasting models using **LSTM, ARIMA, Prophet, TensorFlow, and PyTorch** to forecast product demand and sales trends, reducing forecasting error by **28%**.
- Developed NLP solutions using **SpaCy, NLTK, and Gensim** for sentiment analysis, customer feedback classification, keyword extraction, and topic modeling across customer reviews.
- Engineered scalable data pipelines using **Azure Databricks, PySpark, Hadoop, and Hive**, processing over **100 million** retail transactions monthly while improving ETL efficiency.
- Deployed machine learning models using **MLflow, Docker, Kubernetes, and Azure Machine Learning**, implementing automated model versioning, monitoring, and retraining to reduce deployment time by **70%**.
- Created interactive dashboards and business reports using **Power BI** to visualize KPIs, sales trends, customer behavior, and inventory performance, enabling data-driven decision-making.
- Applied statistical techniques including **hypothesis testing, regression analysis, probability distributions, confidence intervals, and A/B testing** to validate model performance and support business recommendations.
- Optimized SQL queries and database performance across **SQL Server** and **PostgreSQL**, improving reporting efficiency and reducing query execution time by **35%**.

Education

Master of Science in Data Science – University Of Maryland, Baltimore County, USA
Bachelor of Technology in Electronics and Communication Engineering – KL University, Hyderabad, India